

Early Behavioral Identity Formation Predicts Neural Network Training Efficiency

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Abstract

We present evidence that formation score, computed as the Pearson correlation between a model’s confusion matrix at an early checkpoint and its confusion matrix at a mature training checkpoint (here epochs 30 to 50), predicts remaining training need in the tested neural network settings. Models that establish high behavioral identity early converge quickly, while models that establish it late require substantially more training.

The finding rests on a prerequisite observation: in the tested MNIST MLP setting, behavioral identity forms almost immediately during training, reaching 99.46% of its mature value after just one epoch while accuracy was only 82.1%. This suggests that epoch-1 confusion matrices already reflect much of a model’s later confusion structure.

For MLPs, the inverse correlation between formation score and remaining improvement replicated to three decimal places across MNIST and CIFAR-10: $r = -0.780$ and $r = -0.781$ respectively, both with $p < 0.01$ and $n = 7$ architectures. For CNNs, strong inverse correlations were observed on easy and medium-difficulty datasets: $r = -0.987$ on MNIST and $r = -0.978$ on Fashion-MNIST, both with $p < 0.01$ and $n = 8$ architectures. CNN validation on CIFAR-10 did not produce usable correlations at epoch 1 ($r = +0.555$) or epoch 3 ($r = -0.130$), establishing a hard-dataset boundary in the tested CNN setting for the current method.

We also document that formation score does not reliably screen architectures for general quality. In the tested settings, its useful role is estimating remaining training need rather than ranking architectures by final performance. All experiments use real published datasets with no synthetic data. Negative results are reported alongside positive findings.

1 Introduction

Evaluating neural network architectures is expensive (Elsken et al., 2019; Zoph and Le, 2017). The standard approach requires training each candidate architecture to convergence before its efficiency can be assessed. For neural architecture search, hyperparameter optimization, or any workflow that requires comparing many architectures, this creates a substantial computational bottleneck. A method that could estimate training efficiency from early-epoch behavior would allow some slow learners to be identified before full training runs are completed.

This paper presents evidence that such a prediction is possible in the tested settings. We show that a behavioral identity formation score, computed as the Pearson correlation between a model’s confusion matrix at an early checkpoint and its confusion matrix at a mature training checkpoint, is strongly inversely correlated with the remaining improvement a model needs to reach mature behavior in our tested architectures and datasets. Models that form high behavioral identity early

converge quickly, while models that form low behavioral identity early require substantially more training.

The finding rests on a prerequisite observation: in the tested MNIST MLP setting, behavioral identity forms almost immediately during training, well before final accuracy is reached. A model’s confusion structure was 99.46% established after just one epoch of training, even though accuracy was only 82.1% at that point. This suggests that, in the tested setting, epoch-1 confusion matrices already contain predictive information about the rest of training.

We validate the formation-efficiency relationship across two architecture families and four datasets. For MLPs, the inverse correlation between formation score and remaining improvement needed replicated to three decimal places across MNIST and CIFAR-10: $r = -0.780$ and $r = -0.781$ respectively, both with $p < 0.01$ and $n = 7$ architectures. For CNNs, strong inverse correlations were observed on the tested easy and medium-difficulty datasets: $r = -0.987$ on MNIST and $r = -0.978$ on Fashion-MNIST, both with $p < 0.01$ and $n = 8$ architectures. We also document the boundary conditions where the method did not produce usable correlations: CNN validation on CIFAR-10 did not succeed at epoch 1 ($r = +0.555$) or epoch 3 ($r = -0.130$).

This paper makes the following contributions:

- We show that behavioral identity formation is largely complete after one epoch of training, well before accuracy converges, in the tested MLP setting.
- We show that, for the tested MLP architectures, epoch-1 formation score is strongly inversely correlated with remaining improvement on MNIST and CIFAR-10.
- We show that the same relationship was observed for the tested CNN architectures on easy and medium-difficulty datasets, with very strong inverse correlations in our sample.
- We document explicit boundary conditions where the method did not produce usable predictions, including CNN validation on CIFAR-10 at both epoch 1 and epoch 3.
- We show that formation score does not reliably screen architectures for general quality; in these experiments, its useful role is estimating training efficiency rather than ranking architectures by final performance.

All experiments use real published datasets with no synthetic data. We report negative results alongside positive findings.

2 The Formation Score Framework

2.1 Motivation

Standard approaches to neural architecture search and training efficiency estimation require running each candidate architecture for many epochs before meaningful performance differences emerge. This makes architecture comparison expensive, particularly when the number of candidate architectures is large.

The formation score approach rests on a different premise: that the behavioral fingerprint of a trained model is largely established very early in training, even before accuracy has converged. If this is true, then measurements taken after just one epoch of training should carry predictive information about how much additional training the model will need.

2.2 The Formation Score

For a classification model, the confusion matrix captures the model’s observed pattern of correct predictions and errors across the class set. The formation score measures how similar a model’s confusion structure is at an early checkpoint relative to its confusion structure at a mature checkpoint.

Formally, given a model trained to a mature checkpoint producing confusion matrix CM_{mature} , and the same model evaluated after minimal early training producing confusion matrix CM_{early} , the formation score is:

$$\text{Formation Score} = \max(0, \text{Pearson}(\text{flatten}(CM_{early}), \text{flatten}(CM_{mature}))) \quad (1)$$

where both matrices are flattened to one-dimensional vectors before correlation is computed, and the $\max(0, \cdot)$ floor prevents negative correlations from being reported as meaningful anti-similarity.

The result is a score in $[0.0, 1.0]$:

- A score near 1.0 means the model’s confusion structure at the early checkpoint closely matches its mature confusion structure. The model has already established most of its behavioral fingerprint.
- A score near 0.0 means the model’s confusion structure at the early checkpoint bears little resemblance to its mature state. The model has not yet established its behavioral fingerprint.

2.3 What Formation Score Measures and What It Does Not

Formation score measures structural similarity between early and mature confusion patterns. It does not measure accuracy, performance quality, or whether a model is good or bad. A model with a high formation score may be performing well or poorly at the early checkpoint, having simply already established the pattern of errors it will make at maturity.

This distinction matters for interpreting the training efficiency prediction results. The inverse correlation between formation score and remaining improvement is not a claim that high-formation models are better architectures. It is a claim that high-formation models have less behavioral work left to do, because their confusion structure is already close to its mature state.

Formation score also assumes that the mature checkpoint is a reasonable approximation of final model behavior. In the experiments reported here, mature checkpoints were set at epochs 30 to 50 depending on the architecture and dataset.

3 Baseline Validation: Formation Score on a Single Model Across Training

3.1 Setup

Before applying the formation score to efficiency prediction, we first check whether the metric behaves sensibly on a model progressing through normal training. If formation scores were noisy or unstable across training epochs, they would not be useful as a measurement instrument.

We trained a standard MLP on MNIST (LeCun et al., 1998) (7,000 training samples, 3,000 test samples) to a mature checkpoint and computed the formation score at epochs 1, 2, 5, 10, 15, 20, 25, 30, 40, and 50, each measured against the epoch-50 mature checkpoint.

3.2 Results

Epoch	Accuracy	Formation Score
1	0.8213	0.9946
2	0.8727	0.9976
5	0.9087	0.9991
10	0.9203	0.9998
15	0.9217	0.9997
20	0.9257	0.9998
25	0.9290	0.9999
30	0.9313	0.9999
40	0.9303	0.9999
50	0.9303	1.0000

Table 1: Formation score and accuracy across training epochs for a standard MLP on MNIST. Formation score reaches 0.9946 after just one epoch while accuracy is still 82.1%. Formation score reaches 1.000 by epoch 50.

3.3 What This Shows

Two findings emerge from this baseline. First, formation score remains high and stable across training epochs, increasing rapidly toward its mature value without large fluctuations. This suggests it is a reliable measurement instrument rather than a noisy one in the tested setting.

Second, and more importantly, formation score reaches near-maximum values much earlier than accuracy does. After just one epoch, the model’s confusion structure is already 99.46% established relative to its mature state, even though accuracy is only 82.1%. This establishes the prerequisite for training efficiency prediction: the behavioral fingerprint is largely fixed early, which is what makes epoch-1 formation scores informative about the rest of training.

A diagnostic comparison of the epoch-1 and epoch-50 confusion matrices confirmed that the main class-confusion relationships were already present at epoch 1, with later training primarily changing their strength rather than their overall pattern.

We note that this baseline is on a single architecture on a single dataset. Whether formation score is equally well-behaved across all architecture types and datasets is tested in later sections.

4 Formation Score Predicts Training Efficiency in MLP Architectures

4.1 Setup

We test whether the epoch-1 formation score predicts how much additional training an MLP architecture requires to reach mature behavior. Seven MLP architectures ranging from very small (10-5 neurons) to large (256-128 neurons) were trained on MNIST (LeCun et al., 1998) (7,000 training samples, 3,000 test samples). For each architecture, we measured the formation score at epoch 1 against a mature checkpoint at epoch 30 to 50 depending on experiment, and computed the accuracy improvement from epoch 1 to epoch 10 as a proxy for remaining training need. We then

computed the Pearson correlation between formation score and accuracy improvement across the seven architectures.

To test whether this relationship extends beyond a single dataset in the tested MLP setting, we replicated the same protocol on CIFAR-10 (Krizhevsky, 2009) (50,000 training samples, 10,000 test samples, full dataset, no shortcuts) using the same seven MLP architectures.

4.2 Results

Architecture	Formation Score	Improvement	Epochs to 90% test accuracy
Tiny (10-5)	0.167	0.287	>30
Very small (20-10)	0.604	0.284	>30
Medium-small (50-25)	0.868	0.255	10
Small-medium (64-32)	0.931	0.161	10
Standard (128-64)	0.995	0.087	5
Large (256-128)	0.995	0.061	5
Medium-large (200-100)	0.997	0.057	5

Table 2: Formation score, accuracy improvement from epoch 1 to epoch 10, and epochs needed to reach 90% accuracy for seven MLP architectures on MNIST. Pearson correlation between formation score and improvement: $r = -0.780$, $p < 0.01$, $n = 7$.

Architecture	Formation Score	Improvement	Epochs to 40% test accuracy
Tiny (10-5)	0.760	0.090	20
Very small (20-10)	0.895	0.090	5
Large (256-128)	0.941	0.069	1
Medium-large	0.958	0.062	1
Standard (128-64)	0.961	0.064	1
Small-medium (64-32)	0.962	0.065	1
Medium-small (50-25)	0.980	0.075	1

Table 3: Formation score, accuracy improvement, and epochs needed to reach 40% accuracy for seven MLP architectures on CIFAR-10. Pearson correlation between formation score and improvement: $r = -0.781$, $p < 0.01$, $n = 7$.

Because MNIST and CIFAR-10 differ substantially in task difficulty for the tested MLPs, the epoch-count columns use dataset-specific target accuracies: 90% for MNIST and 40% for CIFAR-10. The reported correlations are computed using formation score versus accuracy improvement, with the threshold columns included only to make the training-efficiency pattern more interpretable.

The Pearson correlation between formation score and accuracy improvement was $r = -0.780$ on MNIST and $r = -0.781$ on CIFAR-10, both statistically significant at $p < 0.01$ with $n = 7$ architectures. The difference between the two correlations is 0.001.

4.3 What This Shows

In the tested MLP architectures, epoch-1 formation score is strongly inversely correlated with the remaining improvement needed to reach mature behavior. Architectures that establish high

behavioral identity early require substantially less additional training, while architectures with low early formation require many more epochs to converge.

The near-identical replication across MNIST and CIFAR-10, two datasets that differ substantially in difficulty, suggests this relationship is not specific to a single benchmark in the tested MLP setting. The absolute formation score range shifts upward on CIFAR-10 relative to MNIST (floor rises from 0.167 to 0.760), but the correlation pattern holds.

We note that the sample size is small ($n = 7$ architectures per dataset) and the tested architectures represent a specific family of fully-connected networks. Whether this relationship holds across a broader range of MLP architectures and datasets is not established by these results alone.

5 CNN Validation: Strong Results on Easy and Medium-Difficulty Datasets with a Hard-Dataset Boundary

5.1 Setup

The MLP results establish the formation-efficiency relationship for fully-connected networks. We next test whether the same relationship holds for convolutional neural networks, which differ substantially from MLPs in both architecture and how features are learned during training.

We tested eight CNN architectures ranging from 24K to 1.1M parameters across four experimental conditions: MNIST at epoch 1, Fashion-MNIST at epoch 1, CIFAR-10 at epoch 1, and CIFAR-10 at epoch 3. Architecture depth ranged from 2-layer to 4-layer convolutional stacks. For each condition, we measured the formation score at the designated early checkpoint against an epoch-50 mature checkpoint, and computed the Pearson correlation between formation score and remaining accuracy improvement across the eight architectures.

Fashion-MNIST (Xiao et al., 2017) was selected as a medium-difficulty dataset, harder than MNIST (clothing items are visually similar to each other) but easier than CIFAR-10 (grayscale, single-object images). CIFAR-10 was tested at both epoch 1 and epoch 3 to determine whether extending the early measurement window would resolve any limitations found at epoch 1.

5.2 Results: MNIST CNNs

Architecture	Formation Score	Improvement	Layer Depth
Tiny CNN	0.610	0.558	2 layers
Very Small CNN	0.621	0.576	2 layers
Small CNN	0.617	0.564	2 layers
Medium CNN	0.961	0.215	3 layers
Standard CNN	0.999	0.060	4 layers
Large CNN	0.998	0.065	4 layers
Very Large CNN	0.999	0.038	4 layers
Huge CNN	0.999	0.044	4 layers

Table 4: Formation score and accuracy improvement for eight CNN architectures on MNIST at epoch 1. Pearson correlation: $r = -0.987$, $p < 0.01$, $n = 8$. Three capacity clusters are visible: 2-layer CNNs (formation ≈ 0.61), 3-layer CNN (formation 0.961), 4-layer CNNs (formation ≈ 0.999).

The Pearson correlation between formation score and remaining improvement was $r = -0.987$ on MNIST CNNs. In our sample, this was stronger in magnitude than the corresponding MLP

correlations. The formation score range was 0.610 to 0.999, a spread of 0.389.

A three-cluster pattern was observed in the formation scores, tracking architectural depth. Two-layer CNNs clustered at formation approximately 0.61 and required 55-57% accuracy improvement to reach maturity. The three-layer CNN reached formation 0.961 and required 21% improvement. Four-layer CNNs clustered near 0.999 and required only 3-6% improvement. This suggests that, in the tested MNIST CNN setting, formation score may expose capacity-threshold effects across architecture depth groups that are not visible from parameter count alone.

5.3 Results: Fashion-MNIST CNNs

Architecture	Formation Score	Improvement
Very Small CNN	0.790	0.355
Tiny CNN	0.836	0.309
Small CNN	0.830	0.324
Medium CNN	0.921	0.225
Large CNN	0.935	0.139
Standard CNN	0.955	0.148
Huge CNN	0.954	0.127
Very Large CNN	0.971	0.127

Table 5: Formation score and accuracy improvement for eight CNN architectures on Fashion-MNIST at epoch 1. Pearson correlation: $r = -0.978$, $p < 0.01$, $n = 8$.

The Pearson correlation on Fashion-MNIST was $r = -0.978$, nearly as strong as the MNIST result. The formation score range was 0.790 to 0.971. This result extends the CNN validation from a simple dataset to a medium-difficulty one.

5.4 Results: CIFAR-10 CNNs, A Boundary Condition

Condition	Measurement Epoch	Formation Range	Correlation
CIFAR-10 CNNs	Epoch 1	0.657–0.772	+0.555
CIFAR-10 CNNs	Epoch 3	0.800–0.940	-0.130

Table 6: CNN formation-efficiency correlation on CIFAR-10 at epoch 1 and epoch 3. Neither measurement produced a usable correlation. The positive correlation at epoch 1 is in the wrong direction. The epoch 3 correlation is negative but too weak to support useful prediction in the tested setting.

Neither CIFAR-10 CNN test produced a usable correlation. At epoch 1, the correlation was $r = +0.555$ in the wrong direction. At epoch 3, the correlation was $r = -0.130$, negative in direction but too weak to support useful prediction in the tested setting. The formation score range remained compressed in both cases relative to the MNIST and Fashion-MNIST results.

5.5 What This Shows

In the tested CNN settings, the formation-efficiency relationship holds strongly on easy and medium-difficulty datasets and fails on the hard dataset tested. A plausible explanation for the CIFAR-10

failure is that the tested CNN architectures learn similar low-level visual features in the first few epochs on complex data, compressing formation scores into a narrow range that does not discriminate well between architectures. On simpler datasets, architectural capacity differences manifest earlier in training, producing the wide formation score ranges needed for the correlation to emerge.

The stronger CNN correlations in the tested settings ($r = -0.987$ and $r = -0.978$ vs $r = -0.780$ and $r = -0.781$ for MLPs) suggest that, in this sample, the method separated architectures more sharply for CNNs than for MLPs when it produced usable correlations.

We note that the CIFAR-10 CNN boundary condition is a genuine limitation of the current method in the tested setting rather than a result to be ignored. Extending the measurement to epoch 3 did not resolve it, suggesting that, in the tested settings, extending the measurement from epoch 1 to epoch 3 was not sufficient to recover a useful correlation.

6 Formation Score Does Not Reliably Screen Architectures for General Quality

6.1 Setup

Before the training efficiency prediction result was discovered, the original hypothesis was different: that formation score at epoch 1 could identify bad architectures by producing low scores for architectures that would ultimately underperform. We document this hypothesis and its failure because it defines the boundary of what formation score actually measures.

We tested five MLP architectures on MNIST, including one architecture labeled as good (128-64 neurons), two undersized architectures (10-5 and 20-10 neurons), one oversized architecture (512-256-128 neurons), and one shallow architecture (100 neurons single layer). If formation score screened bad architectures, undersized, oversized, and shallow architectures should all show low formation scores.

6.2 Results

Architecture	Formation Score	Epoch-1 Accuracy
Good (128-64)	0.995	0.821
Tiny (10-5)	0.167	0.217
Small (20-10)	0.604	0.440
Huge (512-256-128)	0.995	0.864
Single layer (100)	0.992	0.813

Table 7: Formation score and epoch-1 accuracy for five MLP architectures in the architecture screening test on MNIST. Only undersized architectures showed low formation scores. Oversized and shallow architectures showed high formation scores despite being part of the bad-architecture test set.

Only the undersized architectures (10-5 and 20-10) showed low formation scores. The oversized architecture (512-256-128) and the shallow single-layer architecture both showed formation scores above 0.99, nearly identical to the good architecture.

6.3 What This Shows

Formation score does not reliably identify bad architectures in general. In these experiments, formation score reflects how quickly a model’s behavioral fingerprint stabilizes relative to its own mature state, not whether that mature state represents good performance. An oversized architecture can establish its mature confusion pattern very quickly, even when that does not support the original bad-architecture screening hypothesis.

The useful role of formation score is estimating remaining training need, not ranking architectures by final quality. This negative result sharpens the claim and prevents misuse of the metric as a generic architecture-quality signal.

7 Discussion, Limitations, and Future Work

7.1 What This Work Shows

The central finding of this paper is that formation score, measured at epoch 1, is strongly inversely correlated with remaining training need in the tested MLP and CNN settings. Models that establish their behavioral fingerprint early converge quickly. Models that establish it late require substantially more training.

The finding rests on a prerequisite result: in the tested MNIST MLP setting, behavioral identity forms almost immediately, reaching 99.46% of its mature value after just one epoch. This suggests that epoch-1 confusion matrices are not merely early noise; in the tested setting, they already reflect much of the model’s later confusion structure at reduced magnitude.

The MLP replication result is the cleanest finding in the paper. The formation-efficiency correlation was $r = -0.780$ on MNIST and $r = -0.781$ on CIFAR-10 for the same seven MLP architectures, identical to three decimal places across datasets that differ substantially in difficulty. This suggests the relationship is not specific to a single benchmark in the tested MLP setting.

The CNN results extend the finding but with a clear boundary. On MNIST and Fashion-MNIST, CNNs produced very strong correlations ($r = -0.987$ and $r = -0.978$). On CIFAR-10, neither epoch 1 nor epoch 3 measurements produced usable correlations. The method does not currently generalize to the tested hard CNN setting.

7.2 Limitations

Small sample sizes. All efficiency correlation results are based on $n = 7$ or $n = 8$ architectures per dataset. The correlations are strong, but the sample sizes are small. Results should be interpreted as preliminary evidence of a pattern, not as a statistically established law.

Single architecture family per test. MLP results are on fully-connected networks only. CNN results are on convolutional networks only. Cross-family comparisons were not tested.

Hard-dataset CNN boundary. The method failed on CIFAR-10 CNNs at both epoch 1 and epoch 3. The boundary condition appears to be related to early feature similarity across architectures on complex data, but this explanation was not formally tested.

Mature checkpoint definition. Formation score depends on a mature checkpoint being a reasonable approximation of final behavior. In these experiments, mature checkpoints were set at epochs 30 to 50. Whether earlier or later mature checkpoints would change the results is not established.

Validated on image classification only. All experiments use image classification benchmarks. Whether early formation predicts training efficiency in NLP, audio, or tabular domains is unknown.

Formation score does not predict final performance. A high formation score means a model converges quickly, not that it converges to a good result. These are different properties and should not be conflated.

7.3 Future Work

The most direct extension is to test the formation-efficiency relationship on a broader range of architectures and datasets, including larger sample sizes and harder benchmarks for CNNs. The CIFAR-10 CNN failure is the most important open problem, and whether it can be resolved by later measurement epochs, different architecture families, or different datasets with similar difficulty remains to be tested.

A second direction is to test the method on non-image domains. The prerequisite result (early formation) was established only on MNIST MLPs. Whether behavioral identity forms equally early in NLP or tabular models is an open question.

A third direction is to characterize the minimum number of architectures needed to reliably estimate the formation-efficiency correlation. The current $n = 7$ and $n = 8$ results are suggestive but small. A systematic study of sample size requirements would make the method more practically useful.

Finally, the three-cluster pattern observed in the MNIST CNN results, where formation scores grouped sharply by architectural depth rather than parameter count, may reflect a more general phenomenon of capacity-threshold transitions in CNN families. Whether this clustering generalizes across datasets and architecture types is worth investigating as a secondary finding.

Code Availability

The formation score implementation and all validation scripts used in this paper are available at <https://github.com/Wise314/identity-formation-detection>. The repository is publicly available.

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Patent Disclosures

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